Southern Wisconsin Fall Turkey Habitat Suitability Model

A GIS-Based Multi-Criteria Habitat Analysis Using ArcGIS Pro | Updated portfolio case study | October 2026

1. Project Overview

I developed a GIS-based habitat suitability model to identify promising fall turkey hunting habitat across southern Wisconsin, focusing on Wisconsin Turkey Management Zone 1. I integrated land cover, hardwood mast potential, agricultural food sources, forest-edge proximity, terrain characteristics, and road accessibility into a spatially continuous suitability surface.

The project demonstrates raster modeling, spatial data integration, terrain analysis, Python-assisted geoprocessing, cartographic production, and environmental decision support. TurkeySuitability_V1 assigns relative scores to landscape cells; these are not probabilities of turkey occurrence.

Parameter	Specification
Software	ArcGIS Pro / Spatial Analyst
Study area	Southern Wisconsin, Turkey Management Zone 1
Coordinate system	NAD83(HARN) / Wisconsin TM (EPSG:3071)
Analysis resolution	10 m
Primary output	TurkeySuitability_V1
Current status	Habitat model and East-area print layout complete; public-land overlay prepared; raster presentation clipping and ArcGIS Online publication in progress

2. Data Acquisition and Preparation

I assembled datasets from federal and Wisconsin agencies. Exact source product vintages, service IDs, and access dates still require metadata verification.

Dataset / provider	Application
WISCLAND 2.0 Level 4 / Wisconsin DNR	Habitat, forest composition, mast potential
USGS 3D Elevation Program	DEM, slope, topographic position
USDA NASS Cropland Data Layer	Agricultural food sources
WisDOT local and major roads	Road proximity and access
Wisconsin DNR turkey zones	Study area boundary
Wisconsin DNR managed lands and other access datasets	Public-land mapping and hunting-access assessment

I standardized outputs to EPSG:3071 and used a 10 m Rasterbase as the snap raster and cell-size reference. The primary processing boundary was StudyArea_Zone1; a rectangular DataClip polygon was subsequently used to manage smaller export extents. WISCLAND source information is approximately 30 m in resolution; aligning it to 10 m does not create new land-cover detail.

3. General Habitat Suitability — Habitat_Base

I assigned general habitat scores to WISCLAND Level 4 classifications, calculated a numeric HAB_SCORE field using Python, and used Spatial Analyst Lookup to generate Habitat_Base. These values describe broad vegetation and habitat structure, not food availability alone.

Land-cover classification	Score
Central Hardwoods; White Oak/Burr Oak; Red Oak; N. Pin/Black Oak; Other Northern Hardwoods	90
Hemlock Hardwoods; Red Maple	85
Mixed Deciduous/Coniferous Forest; Aspen Forest	80
Pasture; Warm-season Grass	60
Cash Grain	40
Developed, Low Intensity	10
Open Water; Developed, High Intensity	0

4. Hardwood Mast Potential — MastBase

I assigned MAST_SCORE values based on WISCLAND forest composition and generated MastBase with Lookup. This estimates potential mast-producing forest types, not annual acorn production.

Forest classification	Mast score
White Oak, Burr Oak	100
Red Oak	90
N. Pin Oak, Black Oak	85
Central Hardwoods	55
Other Northern Hardwoods	40
Other classifications	0

Python scoring logic:

```
scores = {"White Oak, Burr Oak": 100, "Red Oak": 90, "N. Pin Oak, Black Oak": 85,
"Central Hardwoods": 55, "Other Northern Hardwoods": 40}
```

5. Forest-Edge Proximity — EdgeBase

I classified WISCLAND into forest and nonforest cells, including broad-leaved deciduous, coniferous, mixed forest, and forested wetland classes. A 3 × 3-cell focal minimum identified forest cells adjacent to nonforest cells (ForestBoundary). Distance Accumulation generated ForestEdgeDist, which I reclassified into EdgeBase.

Distance to forest boundary	Score
0–150 m	100
150–300 m	80
300–500 m	50

Distance to forest boundary	Score
Over 500 m	10

This initial version measures proximity to all forest boundaries; it does not distinguish forest–agriculture edges from less productive transitions.

6. Terrain Analysis — TerrainBase

I projected and mosaicked USGS elevation tiles into the 10 m TurkeyGISDEM (33,138 columns × 11,311 rows; 32-bit float; vertical units in meters). I derived Slope_Z1 in degrees and calculated TPI_300m as cell elevation minus the mean elevation in a circular 150 m-radius neighborhood. TPI ranged from −53.31 m to +48.66 m, with mean near zero and standard deviation 2.858 m.

Terrain condition	Score
Gentle upper slopes / ridge shoulders	100
Moderate-slope, near-average terrain	90
Pronounced ridges with manageable slope	80
Other gentle/moderate terrain	70
Steeper or lower-position terrain	45
Very steep terrain	20

Nested conditional expressions combining TPI and slope produced TerrainBase. The categories are scouting heuristics, not empirically calibrated wildlife preferences.

7. Agricultural Food Availability — AgFood_Adj

I extracted CDL_Zone1, assigned FOOD_SCORE by crop class, and used Lookup to create AgFoodBase. I calculated distance to forest using ForestMask_INT, SetNull, and Distance Accumulation to generate ForestDist, then discounted crop scores with distance to cover.

Crop / CDL code	Food score
Corn / 1	100
Sorghum / 4	85
Soybeans / 5	80
Winter wheat / 24	75
Grassland/pasture / 176; Other hay / 37	65
Alfalfa / 36	60
Fallow/idle cropland / 61	55

Distance to forest	Food score retained
0–100 m	100%
100–250 m	75%

Distance to forest	Food score retained
250–500 m	40%
Over 500 m	15%

AgFood_Adj emphasizes agricultural feeding opportunities near cover. CDL classification does not establish harvest dates or actual food remaining in a field.

8. Road Pressure and Accessibility — RoadPressureBase

I combined WisDOT local and major roads into Roads_Combined, calculated RoadDistance using Distance Accumulation, and reclassified it into RoadPressureBase. The score balances disturbance near roads with practical access to more remote habitat.

Distance to road	Score
0–100 m	20
100–250 m	50
250–500 m	80
500–1,000 m	100
1,000–2,000 m	80
Over 2,000 m	60

Road proximity is a proxy for disturbance and access, not a direct measure of hunter use or traffic.

9. Multi-Criteria Suitability Model — TurkeySuitability_V1

I combined six component rasters using a weighted linear overlay with the following weights:

Factor	Weight
General habitat	25%
Hardwood mast potential	20%
Forest-edge proximity	15%
Terrain suitability	15%
Agricultural food availability	15%
Road pressure/accessibility	10%

$S = 0.25H + 0.20M + 0.15E + 0.15T + 0.15A + 0.10R$

Raster implementation:

> 0.25 × Habitat_Base + 0.20 × MastBase + 0.15 × EdgeBase + 0.15 × TerrainBase + 0.15 × AgFood_Adj + 0.10 × RoadPressureBase

Final raster statistic	Value
Minimum	6.50

Final raster statistic	Value
Maximum	97.50
Mean	45.72
Standard deviation	13.83

Preliminary suitability class	Score
Very high	73–100
High	60–73
Moderate	46–60
Low	32–46
Very low	Below 32

A visual review found broad spatial patterns consistent with the model design. This is preliminary quality assurance, not validation against observed turkey locations.

10. Public-Land Mapping and Cartographic Output — Updated

Since the original case study, I have prepared a public/huntable-land polygon overlay and produced an East-area print layout titled “Turkey Hunt Suitability Analysis East.” The layout displays huntable-land polygons in green and a cream-to-red Turkey Scouting Report Score overlay, with a displayed score range of 60–97.5, plus a locator map, scale bar, north arrow, and labeled basemap. The layout covers the Lauderdale Lakes–Whitewater area of southern Wisconsin.

The suitability raster and public-land polygons remain separate layers in ArcGIS Pro, allowing independent symbology and future web-map controls. The East layout demonstrates cartographic integration of high-scoring habitat and mapped land-access areas; it does not, by itself, establish that every depicted property is legally open for fall turkey hunting.

Property-level eligibility, seasonal restrictions, access routes, and source metadata must still be checked against authoritative current rules before the map is presented as an access guide. The earlier proposed names PublicHuntingMask and TurkeyPublicHigh remain workflow concepts unless independently confirmed as actual completed datasets.

11. Web Publication and Presentation-Raster Preparation — In Progress

I prepared a dedicated ArcGIS Pro web map with the final suitability raster and public-land layer. The final score raster is a one-band, 32-bit floating-point file-geodatabase raster with 26,389 columns, 5,746 rows, 10 m cells, and an approximately 578.43 MB uncompressed size.

For ArcGIS Online publication, I configured a raster Tile web layer with the ArcGIS Online / Bing Maps / Google Maps tiling scheme, levels of detail 8–14, and local cache generation. The estimated cache size reported in ArcGIS Pro was approximately 6 MB. Analyzer warning 20034 indicated the web layer would use WGS 1984 Web Mercator; no blocking analyzer error was reported in the conversation. Successful publication of this turkey tile layer has not yet been confirmed.

For supporting raster demonstrations, I created a dissolved 500 m buffer around mapped public lands (PublicLands_500m_Buffer) and began clipping WISCLAND to the actual buffer geometry. The first Clip Raster attempt failed because the input raster path was invalid; subsequent processing and verification of NoData outside the buffer were still underway. Other proposed presentation layers include MastBase, ForestMask_INT, CDL, and a

DEM-derived hillshade. Their clipped outputs and publication are not yet confirmed.

The intended Experience Builder application will display the final suitability surface and independently toggleable public-land polygons, with optional supporting rasters, legend, and methodological context. An Experience Builder application and WordPress embed have been planned, not yet completed.

12. Model Limitations and Quality Assurance

Correlated predictors: Hardwood forest classes can score highly in both Habitat_Base and MastBase; sensitivity to alternative weights remains untested.

Edge quality: ForestBoundary treats all forest boundaries alike rather than differentiating adjacent land use.

Seasonality: CDL does not capture harvest status or fall food availability.

Road effects: Road distance is not a direct measure of traffic, hunting pressure, or parking availability.

Spatial resolution: Resampling approximately 30 m land-cover data onto a 10 m grid does not add source detail.

Ground truth: No independent turkey observations or field validation have been used to calibrate the scoring model.

Access: Public ownership or management polygons do not guarantee legal hunting access; rules may vary by property and season.

Metadata and web output: Verify CDL projection, NoData handling, source vintages, public-land eligibility, final tile cache appearance, and actual published URLs before external release.

13. Next Steps

Verify buffered clipping outputs; publish the final suitability tile layer; publish the public-land polygons as a hosted feature layer; assemble the ArcGIS Online web map; and build an Experience Builder application for portfolio presentation. Longer-term analysis could identify contiguous high-scoring patches with Region Group, calculate patch area and property-level summary scores, test alternative factor weights, and compare candidate scouting areas with field observations.

14. Project Significance

This project demonstrates integration of environmental datasets, raster classification, terrain and proximity analysis, reproducible scoring logic, multi-criteria modeling, public-land cartography, and preparation for web GIS delivery. The workflow is transferable to wildlife habitat assessment, recreational planning, and other environmental suitability applications.

Data Sources and References

Wisconsin DNR — WISCLAND 2.0: https://dnr.wisconsin.gov/maps/WISCLAND
USGS — 3D Elevation Program: https://www.usgs.gov/3d-elevation-program
USDA NASS — Cropland Data Layer: https://www.nass.usda.gov/Research_and_Science/Cropland/
WisDOT — Open Data Portal: https://data-wisdot.opendata.arcgis.com/
Wisconsin DNR — Turkey Hunting: https://dnr.wisconsin.gov/topic/hunt/turkey
Wisconsin DNR — GIS Open Data: https://data-wi-dnr.opendata.arcgis.com/
Source product years, exact service IDs, and access dates remain pending metadata verification. Analysis and model development: ArcGIS Pro, 2026.